

MD. LEKHON MIA

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Education

University of Dhaka (DU)

Master of Science in Biomedical Physics and Technology

Dhaka, Bangladesh

May 2026 – Jul 2027

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Industrial and Production Engineering

Dhaka, Bangladesh

Apr 2019 – Jul 2024

CGPA: 3.46 out of 4.00

Research Interests

- Physical Ergonomics & Biomechanics
- Human Factors in AI/Industry 4.0
- Medical Additive Manufacturing
- Cognitive Ergonomics
- Occupational Health & Safety
- AI/ML for Medical Image Analysis

Relevant Coursework

MSc:

- Human Anatomy & Physiology
- Instrumentation (ECG, EMG, EEG)
- Biomedical Signal Processing
- Biophysics
- Medical Imaging (X-Ray, CT, MRI)
- Bioelectromagnetism & Biophotonics

BSc:

- Ergonomics & Safety Management
- Computer Integrated Manufacturing
- Probability and Statistics
- Manufacturing Process
- Supply Chain Management
- Operation Research

Research Experience

Undergraduate Thesis

Jul 2023 – Jul 2024

Fabrication and Characterization of PVA/ZnO Polymer Composite Using Solution Casting Method

- * Fabricated PVA/ZnO composite films using the solution casting technique.
- * Characterized the mechanical properties including tensile strength, elongation at break, and Young's modulus.
- * Investigated the photocatalytic performance of the composite films

Publications

Mia, M. L., Khan, R. R., & Masud, A. K. M. (2025, September). Fabrication and characterization of PVA/TiO₂/JF composite using solution casting method. *Proceedings of the 15th ICME International Conference on Mechanical Engineering*, Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh.

Selected Projects

Ergonomics Workstation Design

Jan 2024 – Jul 2024

CAD 3D Design Project

- * Designed a 3D workstation model in SolidWorks with adjustable chair features, including leg rest and back support
- * Applied human-factor and ergonomic principles to improve workstation usability and reduce physical discomfort

Sewing Line Inspection Table & Chair Design

Jan 2024 – Jul 2024

Internship Project in Garments

- * Developed a CAD model of an ergonomic inspection table for sewing-line operations
- * Applied anthropometric data and range-of-motion analysis to optimize workstation dimensions and user accessibility

DJI Mavic 3 Pro Drone Design

Jan 2024 – Jul 2024

Catia 3D Product Design Project

- * Designed a detailed 3D model of the DJI Mavic 3 Pro drone using CATIA V5
- * Developed parametric parts and assemblies including propeller blades, arms, and camera gimbal system.

Production Efficiency Evaluation using SMV and Output Rate Calculation

Jul–2024

Arena Simulation Project

- * Conducted Standard Minute Value (SMV) study and calculated hourly production output
- * Identified bottlenecks and productivity improvement opportunities to enhance line balancing

Automatic Water Tank Cleaning Machine

Dec 2022–Sept 2023

Hardware Project

- * Designed and manufactured an automated cleaning system capable of cleaning water tank without manual intervention
- * Performed prototype testing, performance evaluation and optimized motor speed to achieve effective cleaning coverage and reduced operation time

Beach Motor Bike 3D Design

Jul–2021

SolidWorks 3D Product Design Project

- * Developed a 3D CAD model of a beach motor bike in SolidWorks.
- * Designed the frame, suspension system, and drivetrain layout to ensure optimal weight distribution and rider comfort

Technical Skills

Simulation Software: Arena, Ansys Mechanical, COMSOL Multiphysics

CAD Software: SolidWorks, AutoCAD, CATIA

Programming Language: Python, Matlab, C, G-Code, HTML

Hardware Skill: Product Design and Development

Miscellaneous: Microsoft office (Word, Excel, Powerpoint), Matplotlib, L^AT_EX

Technical Experience

Industrial Engineer Intern

Jul–2023

Reedisha Knitex Limited

Gazipur, Dhaka

- * Designed an input table considering ergonomic movement
- * Designed floor layout by AutoCAD with 55.5 percent capacity increament and 53 percent space utilization
- * Calculated SMV, Line Balancing and Optimized Production Process

Industrial Engineer Intern

Feb–2024

Bangladesh Industrial Technical Assistance Center (BITAC)

Tejgaon, Dhaka

- * Gained in-depth knowledge about industrial operations including gear manufacturing, CNC lathe, milling, and EDM machining processes
- * Developed an understanding of CAD/CAM integration and the role of automation in modern manufacturing systems

Startup Experience

Team Fresh Express

Feb 2023 – Present

Supply Chain and Logistic Services

- * Developed a logistics model for transporting fresh vegetables from rural farmers to urban retailers
- * Conducted cost–profit analysis, business feasibility studies, and market research to evaluate operational viability
- * Gained hands-on experience in supply chain optimization and agribusiness logistics management.

Co-Curricular Activities

Senior Executive (Blood Donation)

Jul 2023 – Jul 2024

BADHAN BUET Zone, Suhrawardy Hall

Volunteer

Apr 2019 – Jul 2024

Ankur Internation

Achievements and Certifications

Dutch Bangla Bank Merit Scholarship, DBBF

Undergraduate Level

Board Merit Scholarship, Dhaka Board

Undergraduate Level